Astronomical Instruments: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10

Solution 1 — SNR estimate. Return →

The variance is $10000 + 2500 + 100 + 25 = 12625$, so the noise is 112.4 electrons and SNR $= 10000/112.4 = 89.0$. **Check:** the source dominates, so the answer is close to $\sqrt{10000} = 100$.

Solution 2 — Exposure for SNR 5. Return →

With source and sky rates both 100 e⁻/s, SNR $= 100t/\sqrt{200t} = \sqrt{50t}$. Setting this to 5 gives $50t = 25$, so $t = 0.5$ s. **Check:** the sky doubles the variance, so more time is needed than for a source-only measurement.

Solution 3 — Pixel scale. Return →

The scale is $206.265 \times 4/500 = 1.65''$/pixel. **Check:** this is coarse for $1''$ seeing but may be acceptable for wide-field photometry.

Solution 4 — Spectral resolution. Return →

$\Delta\lambda = 500/5000 = 0.1$ nm. The velocity resolution is $c/R = 3.0 \times 10^5/5000 = 60$ km/s. **Check:** higher R means smaller $\Delta\lambda$ and finer velocity resolution.

Solution 5 — Doppler shift. Return →

$v = c\Delta\lambda/\lambda = 3.0 \times 10^5 \times 0.05/500 = 30$ km/s. **Check:** a 0.01% wavelength shift corresponds to 0.01% of the speed of light.

Solution 6 — Calibration purpose. Return →

Bias corrects the electronic zero level; dark corrects thermal signal; flat corrects pixel-to-pixel sensitivity, vignetting and dust shadows. **Check:** calibration removes instrumental structure before scientific measurement.

Solution 7 — Differential photometry. Return →

A nearby comparison star shares much of the same atmospheric transparency, airmass and instrumental response, so dividing by it cancels common trends. It can fail if the comparison is variable, too far away, very different in colour, or affected by different pixel response or crowding. **Check:** differential photometry removes common-mode noise, not all systematics.

Solution 8 — Saturation. Return →

The maximum exposure is $50000/10000 = 5$ s. A shorter exposure may be chosen to stay in the linear regime, avoid blooming, preserve time resolution or allow for seeing changes that concentrate more light into the brightest pixel. **Check:** saturation destroys photometric information.

Solution 9 — Cadence. Return →

You need at least 40 exposures: 20 during the eclipse and 10 on each side. At 60 s each, the minimum **integration** time is 40 minutes. The **elapsed** session must still cover the 2 h eclipse plus 1 h of baseline on each side: about 4 h, or roughly 40 exposures spread through that window. During the eclipse, 20 points require a cadence of at most 6 minutes. Readout, filtering, guiding, calibration and weather gaps can increase the session length. **Check:** integration time and elapsed time are different budgets.

Solution 10 — Instrument choice. Return →

Prioritize stable photometry and sufficient aperture to reach high SNR. High spectral resolution would spread the light too thin for a 0.02 mag measurement unless the star is very bright. A wide field helps find comparison stars but is secondary. Control systematics such as flat-field errors, colour-dependent extinction, focus drift and comparison-star variability. **Check:** 0.02 mag corresponds to a flux change $1 - 10^{-0.4(0.02)} \approx 1.82\%$ (about 2%), so photon noise must be well below that level.